1 Module 8 Lecture 1: First-Order Perturbation Theory and the Feynman-Hellman Theorem

All problems in quantum mechanics cannot be solved exactly, and thus we need approximation methods, such as the variational method and perturbation theory.

1.1 Example: Simple Harmonic Oscillator

We perturb the spring constant of the Simple Harmonic Oscillator (SHO) by δ.

Definition 1. The perturbed Hamiltonian for a Simple Harmonic Oscillator with a perturbed spring constant is given by:

$$\hat{H}(\delta) = \frac{\hat{p}^2}{2m} + \frac{1}{2}k(1+\delta)\hat{x}^2 \tag{1.1}$$

where the unperturbed Hamiltonian is $\hat{H}(0) = \frac{\hat{p}^2}{2m} + \frac{1}{2}k\hat{x}^2$.

We know the exact energy to be:

$$\begin{aligned} E_n(\delta) &= \hbar\omega\sqrt{1+\delta}\left(n + \frac{1}{2}\right) \\ &= E_n(0)\left(1 + \frac{1}{2}\delta - \frac{1}{8}\delta^2 + \frac{1}{16}\delta^3 + \cdots\right) \\ \text{with } \omega &= \sqrt{\frac{k}{m}} \end{aligned} \tag{1.2}$$

We claim that the first order term of this expansion, $\frac{1}{2}\delta$, can be calculated using the variational method. This is computed with respect to the eigenstates of the unperturbed $\hat{H}$.

Example. Applying the variational method to find the first-order correction for the SHO:

$$\begin{aligned} E_n(\delta) &\approx \langle\psi_n^0|\, \hat{H}(\delta)\, |\psi_n^0\rangle \\ &= \langle\psi_n^0|\, \hat{H}(0) + \frac{1}{2}k\delta\hat{x}^2\, |\psi_n^0\rangle \\ &= E_n(0) + \delta\, \langle\psi_n^0|\, \frac{1}{2}k\hat{x}^2\, |\psi_n^0\rangle \end{aligned} \tag{1.3}$$

By the Virial theorem, for a harmonic oscillator, $\left\langle \hat{K} \right\rangle = \left\langle \hat{V} \right\rangle = \frac{1}{2}E$. So, $\langle\psi_n^0|\, \frac{1}{2}k\hat{x}^2\, |\psi_n^0\rangle = \frac{1}{2}E_n(0)$. Thus, using the variational method, we have:

$$E_n(\delta) \approx E_n(0)\left(1 + \frac{\delta}{2}\right) \tag{1.4}$$

This result agrees with the first-order term in the Taylor series expansion shown in (1.2).

1.2 Feynman-Hellman Theorem

We aim to compute the derivative of an energy with respect to some parameter λ.

Theorem 1 (Feynman-Hellman Theorem). Given a Hamiltonian $\hat{H}(\lambda)$ that depends on a parameter λ, and its corresponding normalized eigenstate $|\psi(\lambda)\rangle$ with eigenvalue $E(\lambda)$, the derivative of the energy with respect to λ is given by:

$$\frac{\mathrm{d}E(\lambda)}{\mathrm{d}\lambda} = \langle\psi(\lambda)| \frac{\mathrm{d}\hat{H}(\lambda)}{\mathrm{d}\lambda} |\psi(\lambda)\rangle \tag{1.5}$$

Proof. We start with the time-independent Schrödinger equation and the normalization condition:

$$\begin{gathered}\hat{H}(\lambda)|\psi(\lambda)\rangle = E(\lambda)|\psi(\lambda)\rangle \\ \langle\psi(\lambda)|\psi(\lambda)\rangle = 1\end{gathered} \tag{1.6}$$

We want to compute $\frac{\mathrm{d}E(\lambda)}{\mathrm{d}\lambda}$. We can express $E(\lambda)$ as the expectation value of $\hat{H}(\lambda)$:

$$E(\lambda) = \langle\psi(\lambda)| \hat{H}(\lambda) |\psi(\lambda)\rangle \tag{1.7}$$

Now, we take the derivative with respect to λ:

$$\begin{aligned}\frac{\mathrm{d}E(\lambda)}{\mathrm{d}\lambda} &= \frac{\mathrm{d}}{\mathrm{d}\lambda}\langle\psi(\lambda)| \hat{H}(\lambda) |\psi(\lambda)\rangle \\ &= \left(\frac{\mathrm{d}}{\mathrm{d}\lambda}\langle\psi(\lambda)|\right)\hat{H}(\lambda)|\psi(\lambda)\rangle + \langle\psi(\lambda)|\frac{\mathrm{d}\hat{H}(\lambda)}{\mathrm{d}\lambda}|\psi(\lambda)\rangle + \langle\psi(\lambda)|\hat{H}(\lambda)\left(\frac{\mathrm{d}}{\mathrm{d}\lambda}|\psi(\lambda)\rangle\right)\end{aligned} \tag{1.8}$$

In the first and third terms, we can substitute $\hat{H}(\lambda)|\psi(\lambda)\rangle = E(\lambda)|\psi(\lambda)\rangle$ and $\langle\psi(\lambda)|\hat{H}(\lambda) = E(\lambda)\langle\psi(\lambda)|$:

$$\begin{aligned}&= \left(\frac{\mathrm{d}}{\mathrm{d}\lambda}\langle\psi(\lambda)|\right)E(\lambda)|\psi(\lambda)\rangle + \langle\psi(\lambda)|\frac{\mathrm{d}\hat{H}(\lambda)}{\mathrm{d}\lambda}|\psi(\lambda)\rangle + \langle\psi(\lambda)|E(\lambda)\left(\frac{\mathrm{d}}{\mathrm{d}\lambda}|\psi(\lambda)\rangle\right) \\ &= \langle\psi(\lambda)|\frac{\mathrm{d}\hat{H}(\lambda)}{\mathrm{d}\lambda}|\psi(\lambda)\rangle + E(\lambda)\left(\left(\frac{\mathrm{d}}{\mathrm{d}\lambda}\langle\psi(\lambda)|\right)|\psi(\lambda)\rangle + \langle\psi(\lambda)|\left(\frac{\mathrm{d}}{\mathrm{d}\lambda}|\psi(\lambda)\rangle\right)\right)\end{aligned} \tag{1.9}$$

The terms inside the parenthesis for $E(\lambda)$ can be recognized as the derivative of the normalization condition:

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\langle\psi(\lambda)|\psi(\lambda)\rangle\right) = \left(\frac{\mathrm{d}}{\mathrm{d}\lambda}\langle\psi(\lambda)|\right)|\psi(\lambda)\rangle + \langle\psi(\lambda)|\left(\frac{\mathrm{d}}{\mathrm{d}\lambda}|\psi(\lambda)\rangle\right) \tag{1.10}$$

Since $\langle\psi(\lambda)|\psi(\lambda)\rangle = 1$, its derivative with respect to λ is 0.

$$\begin{aligned}&= \langle\psi(\lambda)|\frac{\mathrm{d}\hat{H}(\lambda)}{\mathrm{d}\lambda}|\psi(\lambda)\rangle + E(\lambda)\left(\frac{\mathrm{d}}{\mathrm{d}\lambda}1\right) \\ &= \langle\psi(\lambda)|\frac{\mathrm{d}\hat{H}(\lambda)}{\mathrm{d}\lambda}|\psi(\lambda)\rangle + E(\lambda)\cdot 0 \\ &= \langle\psi(\lambda)|\frac{\mathrm{d}\hat{H}(\lambda)}{\mathrm{d}\lambda}|\psi(\lambda)\rangle\end{aligned} \tag{1.11}$$

This completes the proof. □

Example. Using the Feynman-Hellman equation, we can show that the variational form of the first-order energy correction is equivalent to the first-order Taylor expansion derived earlier. Let $\hat{H}(\delta) = \hat{H}(0) + \delta\hat{V}$. Then $E_n(\delta)$ can be expanded as a Taylor series around $\delta = 0$:

$$\begin{aligned} E_n(\delta) &\approx E_n(0) + \delta \left.\frac{\mathrm{d}E_n(\delta)}{\mathrm{d}\delta}\right|_{\delta=0} \\ &= E_n(0) + \delta \left\langle \psi_n^0 \right| \left.\frac{\mathrm{d}\hat{H}(\delta)}{\mathrm{d}\delta}\right|_{\delta=0} \left|\psi_n^0\right\rangle \\ &= E_n(0) + \delta \left\langle \psi_n^0 \right| \frac{\mathrm{d}}{\mathrm{d}\delta}\left(\hat{H}(0) + \delta\hat{V}\right) \left|\psi_n^0\right\rangle \\ &= E_n(0) + \delta \left\langle \psi_n^0 \right| \hat{V} \left|\psi_n^0\right\rangle \end{aligned} \tag{1.12}$$

This result, $E_n(0) + \delta \left\langle \psi_n^0 \right| \hat{V} \left|\psi_n^0\right\rangle$, is precisely the first-order energy correction from perturbation theory, which we saw earlier also matched the variational method's result for the SHO.

1.3 Application: Hyperfine Structure of Hydrogen

1.4 Application: Calculating $\langle n,l| \frac{1}{\hat{r}^2} |n,l\rangle$ by Feynman-Hellman

Problem. For hydrogen, the radial Hamiltonian is given by:

$$\hat{H}_l = \frac{\hat{p}_r^2}{2M} + \frac{\hbar^2 l(l+1)}{2M\hat{r}^2} - \frac{e^2}{\hat{r}} \tag{1.13}$$

and $\hat{H}_l |n,l\rangle = E_n |n,l\rangle$. We are also given the energy for the m-th excited state for a generalized parameter λ:

$$E_m(\lambda) = -\frac{e^2}{2a_0(m+\lambda+1)^2} \tag{1.14}$$

Calculate the expectation value $\langle n,l| \frac{1}{\hat{r}^2} |n,l\rangle$ using the Feynman-Hellman Theorem.

Solution. We generalize the Hamiltonian by replacing l with a continuous parameter λ:

$$\hat{H}(\lambda) = \frac{\hat{p}_r^2}{2M} + \frac{\hbar^2 \lambda(\lambda+1)}{2M\hat{r}^2} - \frac{e^2}{\hat{r}} \tag{1.15}$$

Although l is an integer, making derivatives with respect to l seem invalid, the Feynman-Hellman theorem still applies by considering λ as a formal continuous parameter for the purpose of differentiation, and then setting $\lambda = l$ at the end.

Applying the Feynman-Hellman theorem:

$$\begin{aligned} \frac{\mathrm{d}E_m(\lambda)}{\mathrm{d}\lambda} &= \langle n,l| \frac{\mathrm{d}\hat{H}(\lambda)}{\mathrm{d}\lambda} |n,l\rangle \\ \frac{\mathrm{d}}{\mathrm{d}\lambda}\left(-\frac{e^2}{2a_0(m+\lambda+1)^2}\right) &= \langle n,l| \frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{\hat{p}_r^2}{2M} + \frac{\hbar^2\lambda(\lambda+1)}{2M\hat{r}^2} - \frac{e^2}{\hat{r}}\right) |n,l\rangle \\ -\frac{e^2}{2a_0}(-2)(m+\lambda+1)^{-3}(1) &= \langle n,l| \frac{\hbar^2(2\lambda+1)}{2M\hat{r}^2} |n,l\rangle \\ \frac{e^2}{a_0(m+\lambda+1)^3} &= \langle n,l| \frac{\hbar^2(2\lambda+1)}{2M\hat{r}^2} |n,l\rangle \end{aligned} \tag{1.16}$$

Now we set $m + \lambda + 1 = n$ (where n is the principal quantum number for hydrogen, $n = l+1, l+2, \ldots$ and also $\lambda = l$) and substitute into the equation:

$$\frac{e^2}{a_0 n^3} = \langle n, l| \frac{\hbar^2(2l+1)}{2M\hat{r}^2} |n, l\rangle \tag{1.17}$$

$$\frac{e^2}{a_0 n^3} = \frac{\hbar^2(2l+1)}{2M} \langle n, l| \frac{1}{\hat{r}^2} |n, l\rangle$$

Recalling that the Bohr radius $a_0 = \frac{\hbar^2}{Me^2}$, we can substitute $M = \frac{\hbar^2}{a_0 e^2}$:

$$\frac{e^2}{a_0 n^3} = \frac{\hbar^2(2l+1)}{2(\frac{\hbar^2}{a_0 e^2})\hat{r}^2} \langle n, l| \frac{1}{\hat{r}^2} |n, l\rangle$$

$$\frac{e^2}{a_0 n^3} = \frac{a_0 e^2(2l+1)}{2} \langle n, l| \frac{1}{\hat{r}^2} |n, l\rangle$$

Solving for $\langle n, l| \frac{1}{\hat{r}^2} |n, l\rangle$:

$$\boxed{\langle n, l| \frac{1}{\hat{r}^2} |n, l\rangle = \frac{1}{a_0^2 n^3 (l + \frac{1}{2})}} \tag{1.18}$$

□

2 Module 8 Lecture 2: Second Order Perturbation Theory (via derivatives)

There is a general formal technique to determine perturbation theory to an arbitrary order. However, it is complicated so this lecture only covers the second order-derivation.

2.1 Derivation

Using λ as the small parameter, we write $\hat{H}(\lambda) = \hat{H}_0 + \lambda\hat{V}$. The wavefunction is denoted $|\psi_n(\lambda)\rangle$ and the energy $E_n(\lambda)$. We then perform a Taylor series expansion on the energy:

$$E_n(\lambda) = E_n(0) + \lambda \left.\frac{\mathrm{d}E_n(\lambda)}{\mathrm{d}\lambda}\right|_{\lambda=0} + \frac{\lambda^2}{2} \left.\frac{\mathrm{d}^2 E_n(\lambda)}{\mathrm{d}\lambda^2}\right|_{\lambda=0} + \cdots \tag{2.1}$$

We recall the basic properties:

$$E_n(0) = \langle\psi_n| \hat{H}_0 |\psi_n\rangle, \qquad \hat{H}(\lambda)|\psi_n(\lambda)\rangle = E_n(\lambda)|\psi_n(\lambda)\rangle, \qquad \langle\psi_n(\lambda)|\psi_n(\lambda)\rangle = 1 \tag{2.2}$$

From the Feynman-Hellman theorem, we derived the first derivative:

$$\left.\frac{\mathrm{d}E_n(\lambda)}{\mathrm{d}\lambda}\right|_{\lambda=0} = \langle\psi_n| \left.\frac{\mathrm{d}\hat{H}(\lambda)}{\mathrm{d}\lambda}\right|_{\lambda=0} |\psi_n\rangle = \langle\psi_n| \hat{V} |\psi_n\rangle = V_{nn} \tag{2.3}$$

We now aim to calculate the second derivative, $\left.\frac{\mathrm{d}^2 E_n(\lambda)}{\mathrm{d}\lambda^2}\right|_{\lambda=0}$.

Theorem 2 (Second-Order Perturbation Energy Correction). The second-order energy correction for a non-degenerate eigenvalue E_n due to a perturbation $\lambda\hat{V}$ is given by:

$$E_n^{(2)} = \sum_{m\neq n} \frac{|\langle\psi_m|\,\hat{V}\,|\psi_n\rangle|^2}{E_n^{(0)} - E_m^{(0)}} \tag{2.4}$$

where $E_n^{(0)}$ and $|\psi_n\rangle$ are the unperturbed energy and wavefunction respectively.

Proof. We start by differentiating the Feynman-Hellman result (2.3) with respect to λ:

$$\begin{aligned}
\left.\frac{\mathrm{d}^2 E_n(\lambda)}{\mathrm{d}\lambda^2}\right|_{\lambda=0} &= \frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\langle\psi_n(\lambda)|\,\frac{\mathrm{d}\hat{H}(\lambda)}{\mathrm{d}\lambda}\,|\psi_n(\lambda)\rangle\right)\Bigg|_{\lambda=0} \qquad (2.5)\\
&= \left(\frac{\mathrm{d}}{\mathrm{d}\lambda}\langle\psi_n(\lambda)|\right)\left(\frac{\mathrm{d}\hat{H}(\lambda)}{\mathrm{d}\lambda}\right)|\psi_n(\lambda)\rangle\Bigg|_{\lambda=0}\\
&\quad + \langle\psi_n(\lambda)|\left(\frac{\mathrm{d}^2\hat{H}(\lambda)}{\mathrm{d}\lambda^2}\right)|\psi_n(\lambda)\rangle\Bigg|_{\lambda=0}\\
&\quad + \langle\psi_n(\lambda)|\left(\frac{\mathrm{d}\hat{H}(\lambda)}{\mathrm{d}\lambda}\right)\left(\frac{\mathrm{d}}{\mathrm{d}\lambda}|\psi_n(\lambda)\rangle\right)\Bigg|_{\lambda=0}
\end{aligned}$$

Since $\hat{H}(\lambda) = \hat{H}_0 + \lambda\hat{V}$, we have $\frac{\mathrm{d}\hat{H}(\lambda)}{\mathrm{d}\lambda} = \hat{V}$ and $\frac{\mathrm{d}^2\hat{H}(\lambda)}{\mathrm{d}\lambda^2} = 0$. Evaluating at $\lambda = 0$ and replacing $|\psi_n(\lambda)\rangle$ with $|\psi_n\rangle$, we get:

$$\begin{aligned}
\left.\frac{\mathrm{d}^2 E_n(\lambda)}{\mathrm{d}\lambda^2}\right|_{\lambda=0} &= \left(\frac{\mathrm{d}}{\mathrm{d}\lambda}\langle\psi_n(\lambda)|\right)_{\lambda=0}\hat{V}\,|\psi_n\rangle + \langle\psi_n|\,\hat{V}\left(\frac{\mathrm{d}}{\mathrm{d}\lambda}|\psi_n(\lambda)\rangle\right)_{\lambda=0} \qquad (2.6)\\
&= 2\,\mathrm{Re}\left(\langle\psi_n|\,\hat{V}\left(\frac{\mathrm{d}}{\mathrm{d}\lambda}|\psi_n(\lambda)\rangle\right)_{\lambda=0}\right)
\end{aligned}$$

To compute $\left(\frac{\mathrm{d}}{\mathrm{d}\lambda}|\psi_n(\lambda)\rangle\right)_{\lambda=0}$, we differentiate the Schrödinger equation $\hat{H}(\lambda)\,|\psi_n(\lambda)\rangle = E_n(\lambda)\,|\psi_n(\lambda)\rangle$ with respect to λ:

$$\frac{\mathrm{d}\hat{H}(\lambda)}{\mathrm{d}\lambda}|\psi_n(\lambda)\rangle + \hat{H}(\lambda)\frac{\mathrm{d}}{\mathrm{d}\lambda}|\psi_n(\lambda)\rangle = \frac{\mathrm{d}E_n(\lambda)}{\mathrm{d}\lambda}|\psi_n(\lambda)\rangle + E_n(\lambda)\frac{\mathrm{d}}{\mathrm{d}\lambda}|\psi_n(\lambda)\rangle \tag{2.7}$$

At $\lambda = 0$:

$$\hat{V}\,|\psi_n\rangle + \hat{H}_0\left(\frac{\mathrm{d}}{\mathrm{d}\lambda}|\psi_n(\lambda)\rangle\right)_{\lambda=0} = V_{nn}\,|\psi_n\rangle + E_n(0)\left(\frac{\mathrm{d}}{\mathrm{d}\lambda}|\psi_n(\lambda)\rangle\right)_{\lambda=0} \tag{2.8}$$

Rearranging the terms, we get:

$$(\hat{H}_0 - E_n(0))\left(\frac{\mathrm{d}}{\mathrm{d}\lambda}|\psi_n(\lambda)\rangle\right)_{\lambda=0} = (V_{nn} - \hat{V})\,|\psi_n\rangle \tag{2.9}$$

Since $|\psi_n(\lambda)\rangle$ is normalized, $\langle\psi_n(\lambda)|\psi_n(\lambda)\rangle = 1$. Differentiating gives $\left(\frac{\mathrm{d}}{\mathrm{d}\lambda}\langle\psi_n(\lambda)|\right)|\psi_n(\lambda)\rangle + \langle\psi_n(\lambda)|\left(\frac{\mathrm{d}}{\mathrm{d}\lambda}|\psi_n(\lambda)\rangle\right) = 0$. At $\lambda = 0$:

$$\langle\psi_n|\left(\frac{\mathrm{d}}{\mathrm{d}\lambda}|\psi_n(\lambda)\rangle\right)_{\lambda=0} + \left(\frac{\mathrm{d}}{\mathrm{d}\lambda}\langle\psi_n(\lambda)|\right)_{\lambda=0}|\psi_n\rangle = 0 \tag{2.10}$$

This means $\left(\frac{\mathrm{d}}{\mathrm{d}\lambda}|\psi_n(\lambda)\rangle\right)_{\lambda=0}$ is orthogonal to $|\psi_n\rangle$. Let $\left|\psi_n^{(1)}\right\rangle = \left(\frac{\mathrm{d}}{\mathrm{d}\lambda}|\psi_n(\lambda)\rangle\right)_{\lambda=0}$.

For non-degenerate energy eigenvalues, we can express $\left|\psi_n^{(1)}\right\rangle$ as a linear combination of other unperturbed eigenstates $|\psi_m\rangle$ (where $m \neq n$):

$$\left|\psi_n^{(1)}\right\rangle = \sum_{m\neq n} c_m |\psi_m\rangle \tag{2.11}$$

Applying $\langle\psi_k|$ (for $k \neq n$) to (2.9):

$$\begin{aligned}\langle\psi_k|(\hat{H}_0 - E_n(0))\left|\psi_n^{(1)}\right\rangle &= \langle\psi_k|(V_{nn} - \hat{V})|\psi_n\rangle \\ (E_k(0) - E_n(0))\left\langle\psi_k\middle|\psi_n^{(1)}\right\rangle &= \langle\psi_k|V_{nn}|\psi_n\rangle - \langle\psi_k|\hat{V}|\psi_n\rangle \\ (E_k(0) - E_n(0))c_k &= V_{nn}\langle\psi_k|\psi_n\rangle - V_{kn}\end{aligned} \tag{2.12}$$

Since $\langle\psi_k|\psi_n\rangle = 0$ for $k \neq n$, this simplifies to:

$$\begin{aligned}(E_k(0) - E_n(0))c_k &= -V_{kn} \\ c_k &= \frac{V_{kn}}{E_n(0) - E_k(0)}\end{aligned} \tag{2.13}$$

So, the first-order correction to the wavefunction is:

$$\left|\psi_n^{(1)}\right\rangle = \sum_{m\neq n} \frac{V_{mn}}{E_n(0) - E_m(0)}|\psi_m\rangle \tag{2.14}$$

Now substitute this back into (2.6):

$$\begin{aligned}\left.\frac{\mathrm{d}^2E_n(\lambda)}{\mathrm{d}\lambda^2}\right|_{\lambda=0} &= \langle\psi_n|\hat{V}\sum_{m\neq n}\frac{V_{mn}}{E_n(0) - E_m(0)}|\psi_m\rangle + \sum_{m\neq n}\frac{V_{nm}}{E_n(0) - E_m(0)}\langle\psi_m|\hat{V}|\psi_n\rangle \\ &= \sum_{m\neq n}\frac{V_{nm}V_{mn}}{E_n(0) - E_m(0)} + \sum_{m\neq n}\frac{V_{nm}V_{mn}}{E_n(0) - E_m(0)} \\ &= 2\sum_{m\neq n}\frac{V_{nm}V_{mn}}{E_n(0) - E_m(0)} \\ &= 2\sum_{m\neq n}\frac{|V_{mn}|^2}{E_n(0) - E_m(0)}\end{aligned} \tag{2.15}$$

Therefore, the second-order energy correction term in the Taylor expansion ($E_n^{(2)} = \frac{\lambda^2}{2} \left.\frac{\mathrm{d}^2 E_n(\lambda)}{\mathrm{d}\lambda^2}\right|_{\lambda=0}$) is:

$$E_n^{(2)} = \frac{\lambda^2}{2}\left(2\sum_{m\neq n}\frac{|V_{mn}|^2}{E_n(0)-E_m(0)}\right) = \lambda^2\sum_{m\neq n}\frac{|V_{mn}|^2}{E_n(0)-E_m(0)} \tag{2.16}$$

And the total energy up to second order is:

$$E_n(\lambda) = E_n^{(0)} + \lambda V_{nn} + \lambda^2\sum_{m\neq n}\frac{|V_{mn}|^2}{E_n^{(0)}-E_m^{(0)}} + \cdots \tag{2.17}$$

The wavefunction to first order is:

$$\begin{aligned} |\psi_n(\lambda)\rangle &= |\psi_n\rangle + \lambda\left(\frac{\mathrm{d}}{\mathrm{d}\lambda}|\psi_n(\lambda)\rangle\right)_{\lambda=0} + \cdots \\ &= |\psi_n\rangle + \lambda\sum_{m\neq n}\frac{V_{mn}}{E_n(0)-E_m(0)}|\psi_m\rangle + \cdots \end{aligned} \tag{2.18}$$

□

2.2 Example: Simple Harmonic Oscillator

Example. We apply the second-order perturbation theory to the perturbed Hamiltonian operator from Section 1.1:

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}k\hat{x}^2 + \lambda\left(\frac{1}{2}k\hat{x}^2\right) \tag{2.19}$$

Here, the perturbation is $\hat{V} = \frac{1}{2}k\hat{x}^2$.

First, calculate the first-order energy correction V_{nn}:

$$V_{nn} = \langle n|\,\frac{1}{2}k\hat{x}^2\,|n\rangle = \frac{1}{2}\hbar\omega\left(n+\frac{1}{2}\right) \tag{2.20}$$

So, the energy to first order is:

$$E_n(\lambda) = \hbar\omega\left(n+\frac{1}{2}\right) + \lambda\frac{1}{2}\hbar\omega\left(n+\frac{1}{2}\right) + \cdots \tag{2.21}$$

Next, we need the matrix elements $V_{mn} = \langle m|\,\hat{V}\,|n\rangle$:

$$\begin{aligned} V_{mn} &= \langle m|\,\frac{1}{2}k\hat{x}^2\,|n\rangle \\ &= \frac{1}{2}m\omega^2\frac{\hbar}{2m\omega}\langle m|\,(\hat{a}+\hat{a}^\dagger)^2\,|n\rangle \\ &= \frac{\hbar\omega}{4}\langle m|\,\hat{a}^2 + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} + (\hat{a}^\dagger)^2\,|n\rangle \\ &= \frac{\hbar\omega}{4}\left(\sqrt{n(n-1)}\delta_{m,n-2} + (2n+1)\delta_{m,n} + \sqrt{(n+1)(n+2)}\delta_{m,n+2}\right) \end{aligned} \tag{2.22}$$

Note that the $\delta_{m,n}$ term corresponds to V_{nn} and does not contribute to the second-order sum since $m \neq n$. The terms that contribute to the second-order sum are for $m = n-2$ and $m = n+2$.

The second-order energy correction is $\lambda^2 \sum_{m \neq n} \frac{|V_{mn}|^2}{E_n(0)-E_m(0)}$. For $m = n-2$: $V_{n,n-2} = \frac{\hbar\omega}{4}\sqrt{n(n-1)}$. $E_n(0) - E_{n-2}(0) = \hbar\omega(n+\frac{1}{2}) - \hbar\omega(n-2+\frac{1}{2}) = 2\hbar\omega$. For $m = n+2$: $V_{n,n+2} = \frac{\hbar\omega}{4}\sqrt{(n+1)(n+2)}$. $E_n(0) - E_{n+2}(0) = \hbar\omega(n+\frac{1}{2}) - \hbar\omega(n+2+\frac{1}{2}) = -2\hbar\omega$.

Substituting these into the second-order formula:

$$E_n(\lambda) = \hbar\omega\left(n+\frac{1}{2}\right) + \lambda\frac{1}{2}\hbar\omega\left(n+\frac{1}{2}\right) + \lambda^2\left(\frac{|V_{n,n-2}|^2}{E_n(0)-E_{n-2}(0)} + \frac{|V_{n,n+2}|^2}{E_n(0)-E_{n+2}(0)}\right) + \cdots \tag{2.23}$$

$$= \hbar\omega\left(n+\frac{1}{2}\right) + \lambda\frac{1}{2}\hbar\omega\left(n+\frac{1}{2}\right) + \lambda^2\left(\frac{\left(\frac{\hbar\omega}{4}\right)^2 n(n-1)}{2\hbar\omega} + \frac{\left(\frac{\hbar\omega}{4}\right)^2 (n+1)(n+2)}{-2\hbar\omega}\right) + \cdots$$

$$= \hbar\omega\left(n+\frac{1}{2}\right) + \lambda\frac{1}{2}\hbar\omega\left(n+\frac{1}{2}\right) + \lambda^2\frac{(\hbar\omega)^2}{16\cdot 2\hbar\omega}\left(n(n-1) - (n+1)(n+2)\right) + \cdots$$

$$= \hbar\omega\left(n+\frac{1}{2}\right) + \lambda\frac{1}{2}\hbar\omega\left(n+\frac{1}{2}\right) + \lambda^2\frac{\hbar\omega}{32}\left(n^2 - n - (n^2+3n+2)\right) + \cdots$$

$$= \hbar\omega\left(n+\frac{1}{2}\right) + \lambda\frac{1}{2}\hbar\omega\left(n+\frac{1}{2}\right) + \lambda^2\frac{\hbar\omega}{32}\left(-4n-2\right) + \cdots$$

$$= \hbar\omega\left(n+\frac{1}{2}\right) + \lambda\frac{1}{2}\hbar\omega\left(n+\frac{1}{2}\right) - \lambda^2\frac{\hbar\omega}{16}\left(2n+1\right) + \cdots$$

$$= \hbar\omega\left(n+\frac{1}{2}\right) + \lambda\frac{1}{2}\hbar\omega\left(n+\frac{1}{2}\right) - \lambda^2\frac{\hbar\omega}{8}\left(n+\frac{1}{2}\right) + \cdots$$

$$= \hbar\omega\left(n+\frac{1}{2}\right)\left(1 + \frac{\lambda}{2} - \frac{\lambda^2}{8} + \cdots\right)$$

This result precisely matches the Taylor expansion of the exact energy $E_n(\delta) = \hbar\omega\left(n+\frac{1}{2}\right)\sqrt{1+\lambda}$ from (1.2), confirming the validity of the perturbation theory derivation.

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3 Messiah Chapter XVI

3.1 §15 The Hamiltonian and its Resolvent

Note. A convenient way to organize stationary perturbation theory to all orders is to use the resolvent of the Hamiltonian. For a complex parameter z, the resolvent is defined as:

$$\hat{G}(z) := \frac{1}{z-\hat{H}} \tag{3.1}$$

As a function of z, $\hat{G}(z)$ is analytic everywhere except when $z - \hat{H} = 0$, namely at the eigenvalues of $\hat{H}$. For now, we assume the spectrum is purely discrete.

Let $\{E_i\}$ be the eigenvalues and let $\hat{P}_i$ denote the projector onto the eigenspace associated with E_i. We know that $\hat{P}_i = |E_i\rangle\langle E_i|$ for a nondegenerate eigenvalue E_i and normalized eigenvectors $|E_i\rangle$. These projectors satisfy orthogonality and completeness.

Lemma 1 (Orthogonality of Projectors)**.** For distinct eigenvalues E_i and E_j, the projectors $\hat{P}_i$ and $\hat{P}_j$ are orthogonal:

$$\hat{P}_i\hat{P}_j = \delta_{ij}\hat{P}_i \tag{3.2}$$

Proof. For normalized eigenvectors $|E_i\rangle$, $\hat{P}_i = |E_i\rangle\langle E_i|$.

$$\begin{aligned}
\hat{P}_i\hat{P}_j &= (|E_i\rangle\langle E_i|)(|E_j\rangle\langle E_j|) \\
&= |E_i\rangle \langle E_i|E_j\rangle \langle E_j| \\
&= \langle E_j|E_i\rangle |E_i\rangle\langle E_j| \\
&= \delta_{ij} |E_i\rangle\langle E_i| \\
&= \delta_{ij}\hat{P}_i
\end{aligned} \tag{3.3}$$

This shows the orthogonality property. □

Definition 2 (Completeness Relation)**.** The sum of all projectors onto the eigenspaces of a complete set of eigenvalues forms the identity operator:

$$\sum_i \hat{P}_i = \hat{I} \tag{3.4}$$

Lemma 2. The Hamiltonian operator acts on a projector $\hat{P}_i$ such that:

$$\hat{H}\hat{P}_i = E_i\hat{P}_i \tag{3.5}$$

Proof. Consider an arbitrary state $|\psi\rangle$ in the Hilbert space. Let $\hat{P}_i|\psi\rangle = c|E_i\rangle$ for some scalar c, as $\hat{P}_i$ projects onto the eigenspace of E_i. Now, apply $\hat{H}$ to $\hat{P}_i|\psi\rangle$:

$$\begin{aligned}
\hat{H}\hat{P}_i|\psi\rangle &= \hat{H}(c|E_i\rangle) \\
&= c(\hat{H}|E_i\rangle) \\
&= c(E_i|E_i\rangle) && \text{(since } |E_i\rangle \text{ is an eigenstate of } \hat{H}) \\
&= E_i(c|E_i\rangle) \\
&= E_i(\hat{P}_i|\psi\rangle)
\end{aligned} \tag{3.6}$$

Since this relationship holds for every $|\psi\rangle$ in the Hilbert space, we can conclude that $\hat{H}\hat{P}_i = E_i\hat{P}_i$. □

Proposition 1 (Spectral Decomposition of the Resolvent)**.** The resolvent operator $\hat{G}(z)$ can be expressed as a sum over the projectors of the Hamiltonian's eigenspaces:

$$\hat{G}(z) = \sum_i \frac{\hat{P}_i}{z - E_i} \tag{3.7}$$

Proof. From (3.5), we have $\hat{H}\hat{P}_i = E_i\hat{P}_i$. Rearranging this, we get $(z-\hat{H})\hat{P}_i = (z-E_i)\hat{P}_i$.

Left-multiplying by $\hat{G}(z) = (z - \hat{H})^{-1}$:

$$\begin{aligned} \hat{G}(z)(z-\hat{H})\hat{P}_i &= \hat{G}(z)(z-E_i)\hat{P}_i \\ \hat{I}\hat{P}_i &= \hat{G}(z)(z-E_i)\hat{P}_i \\ \hat{P}_i &= (z-E_i)\hat{G}(z)\hat{P}_i \end{aligned} \tag{3.8}$$

This implies that $\hat{G}(z)\hat{P}_i = \frac{\hat{P}_i}{z-E_i}$. Summing over all i and using the completeness relation (3.4):

$$\begin{aligned} \sum_i \hat{G}(z)\hat{P}_i &= \sum_i \frac{\hat{P}_i}{z-E_i} \\ \hat{G}(z)\sum_i \hat{P}_i &= \sum_i \frac{\hat{P}_i}{z-E_i} \\ \hat{G}(z)\hat{I} &= \sum_i \frac{\hat{P}_i}{z-E_i} \\ \hat{G}(z) &= \sum_i \frac{\hat{P}_i}{z-E_i} \end{aligned} \tag{3.9}$$

This formula shows that each discrete eigenvalue E_i is a simple pole of $\hat{G}(z)$. □

Theorem 3 (Residue of the Resolvent)**.** The residue of the resolvent $\hat{G}(z)$ at a simple pole E_i is the corresponding projector $\hat{P}_i$.

$$\frac{1}{2\pi i}\oint_{\Gamma_i} \hat{G}(z)\mathrm{d}z = \hat{P}_i \tag{3.10}$$

where Γ_i is a small contour enclosing only E_i.

Proof. Using the spectral decomposition (3.7), we can calculate the residue directly:

$$\begin{aligned} \frac{1}{2\pi i}\oint_{\Gamma_i} \hat{G}(z)\mathrm{d}z &= \frac{1}{2\pi i}\oint_{\Gamma_i} \sum_j \frac{\hat{P}_j}{z-E_j}\mathrm{d}z \\ &= \sum_j \frac{1}{2\pi i}\oint_{\Gamma_i} \frac{\hat{P}_j}{z-E_j}\mathrm{d}z \end{aligned} \tag{3.11}$$

By Cauchy's residue theorem, only the term where E_j is inside Γ_i contributes. Since Γ_i only encloses E_i:

$$\begin{aligned} &= \frac{1}{2\pi i}\oint_{\Gamma_i} \frac{\hat{P}_i}{z-E_i}\mathrm{d}z \\ &= \hat{P}_i \cdot 1 \\ &= \hat{P}_i \end{aligned} \tag{3.12}$$

This result is significant as it provides a way to extract projectors using contour integrals. □

Theorem 4 (Projector for a Set of Eigenvalues)**.** For any closed contour Γ that encloses a set of eigenvalues $\{E_j\}$ and excludes all others, the sum of the associated projectors, $\hat{P}_\Gamma$, is given by:

$$\hat{P}_\Gamma = \frac{1}{2\pi i} \oint_\Gamma \hat{G}(z)\mathrm{d}z \tag{3.13}$$

Furthermore, multiplying by $\hat{H}$:

$$\hat{H}\hat{P}_\Gamma = \frac{1}{2\pi i} \oint_\Gamma z\hat{G}(z)\mathrm{d}z \tag{3.14}$$

Proof. The first part follows directly from summing the residues for all E_j inside Γ. For the second part, using $(z - \hat{H})\hat{G}(z) = \hat{I}$:

$$\begin{aligned}
\hat{H}\hat{P}_\Gamma &= \hat{H}\left(\frac{1}{2\pi i} \oint_\Gamma \hat{G}(z)\mathrm{d}z\right) && (3.15)\\
&= \frac{1}{2\pi i} \oint_\Gamma \hat{H}\hat{G}(z)\mathrm{d}z \\
&= \frac{1}{2\pi i} \oint_\Gamma (z\hat{G}(z) - \hat{I})\mathrm{d}z && (\text{since } \hat{H}\hat{G}(z) = z\hat{G}(z) - \hat{I})\\
&= \frac{1}{2\pi i} \oint_\Gamma z\hat{G}(z)\mathrm{d}z - \frac{1}{2\pi i} \oint_\Gamma \hat{I}\mathrm{d}z \\
&= \frac{1}{2\pi i} \oint_\Gamma z\hat{G}(z)\mathrm{d}z - 0 \\
&= \frac{1}{2\pi i} \oint_\Gamma z\hat{G}(z)\mathrm{d}z
\end{aligned}$$

The integral of $\hat{I}$ over a closed contour is zero, assuming the contour is finite and well-behaved. ☐

Theorem 5 (Norm of the Resolvent)**.** The norm of the resolvent operator is given by the reciprocal of the distance from z to the closest eigenvalue:

$$\|\hat{G}(z)\| = \frac{1}{\Delta(z)} \tag{3.16}$$

where $\Delta(z) = \min_i |z - E_i|$ is the distance from z to the closest eigenvalue E_i.

Proof. For any state $|\psi\rangle = \sum_i |\psi_i\rangle$ with $|\psi_i\rangle = \hat{P}_i |\psi\rangle$, we can write:

$$\begin{aligned}
\|\hat{G}(z)\,|\psi\rangle\,\|^2 &= \left\|\sum_i \frac{\hat{P}_i\,|\psi\rangle}{z - E_i}\right\|^2 && (3.17)\\
&= \left\|\sum_i \frac{|\psi_i\rangle}{z - E_i}\right\|^2
\end{aligned}$$

Since the $|\psi_i\rangle$ components are orthogonal (as they lie in orthogonal eigenspaces), the norm squared is:

$$
\begin{aligned}
&= \sum_i \left\| \frac{|\psi_i\rangle}{z - E_i} \right\|^2 \\
&= \sum_i \frac{\| \, |\psi_i\rangle \, \|^2}{|z - E_i|^2} \\
&\leq \left(\max_i \frac{1}{|z - E_i|^2} \right) \sum_i \| \, |\psi_i\rangle \, \|^2 \\
&= \left(\max_i \frac{1}{|z - E_i|^2} \right) \| \, |\psi\rangle \, \|^2
\end{aligned}
$$

Taking the square root, we get $\|\hat{G}(z)\| \leq \max_i \frac{1}{|z-E_i|} = \frac{1}{\min_i |z-E_i|} = \frac{1}{\Delta(z)}$.

To show equality, choose a state $|\psi\rangle$ that lies entirely within an eigenspace corresponding to an eigenvalue E_k such that $|z - E_k| = \Delta(z)$. For such a state, $\hat{G}(z) \, |\psi\rangle = \frac{1}{z-E_k} \, |\psi\rangle$. Then, $\|\hat{G}(z) \, |\psi\rangle \, \| = \left| \frac{1}{z-E_k} \right| \| \, |\psi\rangle \, \| = \frac{1}{\Delta(z)} \| \, |\psi\rangle \, \|$. Therefore, the operator norm is exactly $\frac{1}{\Delta(z)}$. □

3.2 §16 Expansion of $\hat{G}(z)$, $\hat{P}$ and $\hat{H}\hat{P}$ in powers of $\lambda\hat{V}$

Definition 3 (Perturbed and Unperturbed Resolvents). We write the perturbed Hamiltonian as $\hat{H} = \hat{H}_0 + \lambda\hat{V}$, where λ is a bookkeeping parameter. The unperturbed and full resolvents are defined as:

$$\hat{G}_0(z) = \frac{1}{z - \hat{H}_0} \tag{3.18}$$

$$\hat{G}(z) = \frac{1}{z - \hat{H}_0 - \lambda\hat{V}} \tag{3.19}$$

Proposition 2 (Equation for Resolvents). The full resolvent $\hat{G}(z)$ can be expressed in terms of the unperturbed resolvent $\hat{G}_0(z)$ and the perturbation $\hat{V}$ as:

$$\hat{G} = \hat{G}_0(z)(1 + \lambda\hat{V}\hat{G}(z)) \tag{3.20}$$

Proof. We start with the definition of $\hat{G}(z)$:

$$\begin{aligned}\hat{G}(z) &= \frac{1}{z-\hat{H}_0-\lambda\hat{V}} \\ &= \frac{1}{z-\hat{H}_0-\lambda\hat{V}} \\ &= \frac{1}{z-\hat{H}_0}(z-\hat{H}_0)\frac{1}{z-\hat{H}_0-\lambda\hat{V}} \\ &= \frac{1}{z-\hat{H}_0}[(z-\hat{H}_0-\lambda\hat{V})+\lambda\hat{V}]\frac{1}{z-\hat{H}_0-\lambda\hat{V}} \\ &= \frac{1}{z-\hat{H}_0}(z-\hat{H}_0-\lambda\hat{V})\frac{1}{z-\hat{H}_0-\lambda\hat{V}}+\frac{1}{z-\hat{H}_0}\lambda\hat{V}\frac{1}{z-\hat{H}_0-\lambda\hat{V}} \\ &= \frac{1}{z-\hat{H}_0}\hat{I}+\frac{1}{z-\hat{H}_0}\lambda\hat{V}\hat{G}(z) \\ &= \hat{G}_0(z)+\hat{G}_0(z)\lambda\hat{V}\hat{G}(z) \\ &= \hat{G}_0(z)(\hat{I}+\lambda\hat{V}\hat{G}(z))\end{aligned} \tag{3.21}$$

□

Theorem 6 (Neumann Series Expansion of the Resolvent)**.** The full resolvent $\hat{G}(z)$ can be expressed as a Neumann series:

$$\hat{G}(z) = \sum_{n=0}^{\infty} \lambda^n \hat{G}_0(z)(\hat{V}\hat{G}_0(z))^n \tag{3.22}$$

This series converges when $\|\lambda\hat{V}\hat{G}_0(z)\| < 1$.

Proof. From Proposition 2, we have $\hat{G} = \hat{G}_0 + \hat{G}_0\lambda\hat{V}\hat{G}$. We can rearrange this to isolate $\hat{G}$:

$$\begin{aligned}\hat{G}-\hat{G}_0\lambda\hat{V}\hat{G} &= \hat{G}_0 \\ (\hat{I}-\hat{G}_0\lambda\hat{V})\hat{G} &= \hat{G}_0 \\ \hat{G} &= (\hat{I}-\hat{G}_0\lambda\hat{V})^{-1}\hat{G}_0\end{aligned} \tag{3.23}$$

Using the geometric series expansion for operators, $(\hat{I}-\hat{X})^{-1} = \sum_{n=0}^{\infty}\hat{X}^n$, which converges if the operator norm $\|\hat{X}\| < 1$. Here, $\hat{X} = \hat{G}_0\lambda\hat{V}$. So, the series converges if $\|\hat{G}_0\lambda\hat{V}\| < 1$.

$$\begin{aligned}\hat{G} &= \left(\sum_{n=0}^{\infty}(\hat{G}_0\lambda\hat{V})^n\right)\hat{G}_0 \\ &= \sum_{n=0}^{\infty}\lambda^n(\hat{G}_0\hat{V})^n\hat{G}_0 \\ &= \sum_{n=0}^{\infty}\lambda^n\hat{G}_0(\hat{V}\hat{G}_0)^n\end{aligned} \tag{3.24}$$

The convergence condition is $\|\lambda\hat{V}\hat{G}_0\| < 1$. Since $\|\hat{G}_0(z)\| = 1/\Delta_0(z)$ (where $\Delta_0(z)$ is the distance from z to the nearest eigenvalue of $\hat{H}_0$), a sufficient condition for convergence is $\|\lambda\hat{V}\| < \Delta_0(z)$. This means that the perturbation must be sufficiently small. □

Proposition 3 (Expansion of the Projector $\hat{P}$)**.** The projector $\hat{P}$ onto an eigenspace of the perturbed Hamiltonian can be expanded as a power series in the perturbation parameter λ:

$$\hat{P} = \hat{P}_0 + \sum_{n=1}^{\infty} \lambda^n \hat{A}^{(n)} \tag{3.25}$$

where $\hat{P}_0$ is the unperturbed projector and $\hat{A}^{(n)} = \frac{1}{2\pi i}\oint_{\Gamma_a} \hat{G}_0(\hat{V}\hat{G}_0)^n \mathrm{d}z$.

Proof. Let Γ_a be a contour that encloses the unperturbed eigenvalue E_a^0 and the eigenvalues of $\hat{H}$ that tend to E_a^0 for small λ. From the residue formula (3.13), the projector $\hat{P}$ is:

$$\hat{P} = \frac{1}{2\pi i}\oint_{\Gamma_a} \hat{G}(z)\mathrm{d}z \tag{3.26}$$

Substituting the Neumann series for $\hat{G}(z)$ from (3.22):

$$\begin{aligned} \hat{P} &= \frac{1}{2\pi i}\oint_{\Gamma_a} \sum_{n=0}^{\infty} \lambda^n \hat{G}_0(z)(\hat{V}\hat{G}_0(z))^n \mathrm{d}z \qquad (3.27)\\ &= \frac{1}{2\pi i}\oint_{\Gamma_a} \hat{G}_0(z)\mathrm{d}z + \sum_{n=1}^{\infty} \lambda^n \left(\frac{1}{2\pi i}\oint_{\Gamma_a} \hat{G}_0(z)(\hat{V}\hat{G}_0(z))^n \mathrm{d}z\right) \end{aligned}$$

The first term is the projector onto the unperturbed eigenspace corresponding to E_a^0, denoted $\hat{P}_0$. The remaining terms define $\hat{A}^{(n)}$:

$$\hat{P} = \hat{P}_0 + \sum_{n=1}^{\infty} \lambda^n \hat{A}^{(n)} \tag{3.28}$$

□

To evaluate the residues for $\hat{A}^{(n)}$, we expand $\hat{G}_0(z)$ about E_a^0 in a Laurent series. Let $\hat{Q}_a := \hat{I} - \hat{P}_a = \sum_{i\neq a} \hat{P}_i$, which is the projector onto the subspace orthogonal to the unperturbed eigenspace of E_a^0.

Definition 4 (Laurent Series of Unperturbed Resolvent)**.** The unperturbed resolvent $\hat{G}_0(z)$ can be written in terms of projectors:

$$\hat{G}_0(z) = \frac{\hat{P}_a}{z - E_a^0} + \sum_{i\neq a} \frac{\hat{P}_i}{z - E_i^0} \tag{3.29}$$

For $|z - E_a^0| < \min_{i\neq a} |E_i^0 - E_a^0|$, we can expand the second term as a geometric series.

Expanding a single term $\frac{1}{z-E_i^0}$:

$$\frac{1}{z-E_i^0} = \frac{1}{(z-E_a^0)-(E_i^0-E_a^0)} \tag{3.30}$$

$$= \frac{1}{E_i^0-E_a^0}\frac{E_i^0-E_a^0}{(z-E_a^0)-(E_i^0-E_a^0)} \tag{3.31}$$

$$= -\frac{1}{E_i^0-E_a^0}\frac{1}{1-\frac{z-E_a^0}{E_i^0-E_a^0}} \tag{3.32}$$

$$= -\sum_{k=0}^{\infty}\frac{(z-E_a^0)^k}{(E_i^0-E_a^0)^{k+1}} \tag{3.33}$$

The convergence condition for this geometric series is $\left|\frac{z-E_a^0}{E_i^0-E_a^0}\right| < 1$. Summing over $i \neq a$:

$$\sum_{i\neq a}\frac{\hat{P}_i}{z-E_i^0} = \sum_{k=0}^{\infty}(-1)^k(z-E_a^0)^k\sum_{i\neq a}\frac{\hat{P}_i}{(E_a^0-E_i^0)^{k+1}} \tag{3.34}$$

Definition 5 (Operator $\hat{O}_a$)**.** Define the operator $\hat{O}_a$, which is the inverse of $(E_a^0-\hat{H}_0)$ on the $\hat{Q}_a$ subspace:

$$\hat{O}_a := \hat{Q}_a(E_a^0-\hat{H}_0)^{-1}\hat{Q}_a \tag{3.35}$$

Lemma 3 (Explicit Form of $\hat{O}_a$)**.** The operator $\hat{O}_a$ can be explicitly written as:

$$\hat{O}_a = \sum_{i\neq a}\frac{1}{E_a^0-E_i^0}\hat{P}_i \tag{3.36}$$

Proof. Consider the operator $\hat{X} = \sum_{i\neq a}\frac{1}{E_a^0-E_i^0}\hat{P}_i$. We need to show that this is the inverse of $(E_0^a-\hat{H}_0)$ on the $\hat{Q}_a$ subspace. This means we need to show $(E_0^a-\hat{H}_0)\hat{X} = \hat{Q}_a$ and $\hat{X}(E_0^a-\hat{H}_0) = \hat{Q}_a$.

First, consider $(E_0^a-\hat{H}_0)\hat{X}$:

$$(E_0^a-\hat{H}_0)\hat{X} = (E_a^0-\hat{H}_0)\left(\sum_{i\neq a}\frac{\hat{P}_i}{E_a^0-E_i^0}\right) \tag{3.37}$$

$$= \left(\sum_j (E_a^0-E_j^0)\hat{P}_j\right)\left(\sum_{i\neq a}\frac{\hat{P}_i}{E_a^0-E_i^0}\right)$$

Using the orthogonality of projectors $\hat{P}_j\hat{P}_i = \delta_{ji}\hat{P}_i$:

$$
\begin{aligned}
&= \sum_j \sum_{i\neq a} \frac{(E_a^0 - E_j^0)}{(E_a^0 - E_i^0)} \hat{P}_j \hat{P}_i \\
&= \sum_{i\neq a} \frac{(E_a^0 - E_i^0)}{(E_a^0 - E_i^0)} \hat{P}_i \\
&= \sum_{i\neq a} \hat{P}_i \\
&= \hat{Q}_a
\end{aligned}
$$

Now, consider $\hat{X}(E_0^a - \hat{H}_0)$:

$$
\begin{aligned}
\hat{X}(E_0^a - \hat{H}_0) &= \left(\sum_{i\neq a} \frac{\hat{P}_i}{E_a^0 - E_i^0} \right) \left(\sum_j (E_a^0 - E_j^0) \hat{P}_j \right) \qquad (3.38) \\
&= \sum_{i\neq a} \sum_j \frac{(E_a^0 - E_j^0)}{(E_a^0 - E_i^0)} \hat{P}_i \hat{P}_j \\
&= \sum_{i\neq a} \frac{(E_a^0 - E_i^0)}{(E_a^0 - E_i^0)} \hat{P}_i \\
&= \sum_{i\neq a} \hat{P}_i \\
&= \hat{Q}_a
\end{aligned}
$$

Thus, we have shown that $\hat{X}$ is the left and right inverse of $(E_0^a - \hat{H}_0)$ in the $\hat{Q}_a$ subspace, which is precisely the definition of $\hat{O}_a$. Therefore, $\hat{O}_a = \sum_{i\neq a} \frac{1}{E_a^0 - E_i^0} \hat{P}_i$. □

Lemma 4 (Powers of $\hat{O}_a$). The k-th power of $\hat{O}_a$ is given by:

$$
(\hat{O}_a)^k = \sum_{i\neq a} \frac{\hat{P}_i}{(E_a^0 - E_i^0)^k} \qquad (3.39)
$$

Proof. This can be shown by induction. For $k = 1$, it is the definition of $\hat{O}_a$. Assuming it holds for k, we show for $k+1$:

$$
\begin{aligned}
(\hat{O}_a)^{k+1} &= (\hat{O}_a)^k \hat{O}_a && (3.40)\\
&= \left(\sum_{j\neq a} \frac{\hat{P}_j}{(E_a^0 - E_j^0)^k}\right)\left(\sum_{i\neq a} \frac{\hat{P}_i}{E_a^0 - E_i^0}\right)\\
&= \sum_{j\neq a}\sum_{i\neq a} \frac{\hat{P}_j \hat{P}_i}{(E_a^0 - E_j^0)^k (E_a^0 - E_i^0)}\\
&= \sum_{i\neq a} \frac{\hat{P}_i}{(E_a^0 - E_i^0)^k (E_a^0 - E_i^0)} && (\text{since } \hat{P}_j \hat{P}_i = \delta_{ji}\hat{P}_i)\\
&= \sum_{i\neq a} \frac{\hat{P}_i}{(E_a^0 - E_i^0)^{k+1}}
\end{aligned}
$$

This completes the proof. □

Substituting (3.39) into (3.34), we can simplify the Laurent series for the unperturbed resolvent:

$$
\begin{aligned}
\hat{G}_0(z) &= \frac{\hat{P}_a}{z - E_a^0} + \sum_{k=0}^{\infty} (-1)^k (z - E_a^0)^k \sum_{i\neq a} \frac{\hat{P}_i}{(E_a^0 - E_i^0)^{k+1}} && (3.41)\\
&= \frac{\hat{P}_a}{z - E_a^0} + \sum_{k=0}^{\infty} (-1)^k (z - E_a^0)^k (\hat{O}_a)^{k+1}\\
&= \frac{\hat{P}_a}{z - E_a^0} - \sum_{k=1}^{\infty} (-1)^{k-1} (z - E_a^0)^{k-1} (\hat{O}_a)^k\\
&= \frac{\hat{P}_a}{z - E_a^0} + \sum_{k=1}^{\infty} (-1)^{k-1} (z - E_a^0)^{k-1} \hat{O}_a^k
\end{aligned}
$$

This expression is a Laurent series where the principal part corresponds to the projector $\hat{P}_a$ and the analytic part involves powers of $\hat{O}_a$.

Definition 6 (Kato's Operators $\hat{S}^k$)**.** Kato's operators $\hat{S}^k$ are defined as:

$$
\hat{S}^k = \begin{cases} -\hat{P}_a, & k = 0, \\ \hat{O}_a^k, & k \geq 1. \end{cases} \tag{3.42}
$$

Using these operators, the Laurent expansion of $\hat{G}_0(z)$ around E_a^0 can be written compactly as:

$$
\hat{G}_0(z) = \sum_{k=0}^{\infty} (-1)^{k-1} (z - E_a^0)^{k-1} \hat{S}^k \tag{3.43}
$$

Multiplying both sides by $\hat{V}$,

$$
\hat{V}\hat{G}_0 = \sum_{k=0}^{\infty} (-1)^{k-1} (z - E_a^0)^{k-1} \hat{V}\hat{S}^k \tag{3.44}
$$

Lemma 5 (Powers of $\hat{V}\hat{G}_0$)**.** The nth power of $\hat{V}\hat{G}_0$ is given by

$$(\hat{V}\hat{G}_0)^n = \sum_{k_1,\ldots,k_n \geq 0} \left[\prod_{j=1}^{n} (-1)^{k_j-1}(z - E_0^0)^{k_j-1} \right] (\hat{V}\hat{S}^{k_1}) \cdots (\hat{V}\hat{S}^{k_n}). \tag{3.45}$$

Proof. We prove by induction on $n \geq 1$ that

$$(\hat{V}\hat{G}_0)^n = \sum_{k_1,\ldots,k_n \geq 0} \left[\prod_{j=1}^{n} (-1)^{k_j-1}(z - E_0^0)^{k_j-1} \right] (\hat{V}\hat{S}^{k_1}) \cdots (\hat{V}\hat{S}^{k_n}).$$

Base case $n = 1$*.* This is exactly the displayed formula for $\hat{V}\hat{G}_0$.

Induction step. Assume $(*)$ holds for n. Then

$$\begin{aligned}(\hat{V}\hat{G}_0)^{n+1} &= (\hat{V}\hat{G}_0)^n(\hat{V}\hat{G}_0) \\ &= \left[\sum_{k_1,\ldots,k_n \geq 0} \Big(\prod_{j=1}^{n} (-1)^{k_j-1}(z - E_0^0)^{k_j-1} \Big) (\hat{V}\hat{S}^{k_1}) \cdots (\hat{V}\hat{S}^{k_n}) \right] \left[\sum_{k_{n+1} \geq 0} (-1)^{k_{n+1}-1}(z - E_0^0)^{k_{n+1}-1} \hat{V}\hat{S}\right.\end{aligned}$$

Expanding the product of sums (choose one term from each sum, preserving operator order) gives

$$(\hat{V}\hat{G}_0)^{n+1} = \sum_{k_1,\ldots,k_{n+1} \geq 0} \Big(\prod_{j=1}^{n+1} (-1)^{k_j-1}(z - E_0^0)^{k_j-1} \Big) (\hat{V}\hat{S}^{k_1}) \cdots (\hat{V}\hat{S}^{k_{n+1}}),$$

since the bracketed factors are scalars (functions of z) and hence commute. This is $(*)$ with $n \mapsto n+1$. By induction, the identity holds for all $n \geq 1$. □

Now we use this compact form to expand $(\hat{V}\hat{G}_0)^n$:

$$(\hat{V}\hat{G}_0)^n = \prod_{j=1}^{n} \left(\hat{V} \sum_{k_j \geq 0} (-1)^{k_j-1}(z - E_a^0)^{k_j-1} \hat{S}^{k_j} \right) \tag{3.46}$$

$$= \prod_{j=1}^{n} \left(\sum_{k_j \geq 0} (-1)^{k_j-1}(z - E_a^0)^{k_j-1} \hat{V}\hat{S}^{k_j} \right) \tag{3.47}$$

$$= \sum_{k_1,\cdots,k_n \geq 0} \left[\prod_{j=1}^{n} (-1)^{k_j-1}(z - E_a^0)^{k_j-1} \right] (\hat{V}\hat{S}^{k_1}) \cdots (\hat{V}\hat{S}^{k_n}) \tag{3.48}$$

$$= \sum_{k_1,\cdots,k_n \geq 0} (-1)^{\left(\sum_{j=1}^{n} k_j\right)-n} (z - E_a^0)^{\left(\sum_{j=1}^{n} k_j\right)-n} (\hat{V}\hat{S}^{k_1}) \cdots (\hat{V}\hat{S}^{k_n}) \tag{3.49}$$

Finally, we combine $\hat{G}_0$ with $(\hat{V}\hat{G}_0)^n$:

$$\hat{G}_0(\hat{V}\hat{G}_0)^n = \left[\sum_{k_0\geq 0}(-1)^{k_0-1}(z-E_a^0)^{k_0-1}\hat{S}^{k_0}\right]\times \tag{3.50}$$

$$\left[\sum_{k_1,\cdots,k_n\geq 0}(-1)^{\left(\sum_{j=1}^n k_j\right)-n}(z-E_a^0)^{\left(\sum_{j=1}^n k_j\right)-n}(\hat{V}\hat{S}^{k_1})\cdots(\hat{V}\hat{S}^{k_n})\right] \tag{3.51}$$

$$= \sum_{k_0,\cdots,k_n\geq 0}(-1)^{(k_0-1)+\left(\sum_{j=1}^n k_j\right)-n}(z-E_a^0)^{(k_0-1)+\left(\sum_{j=1}^n k_j\right)-n}\left(\hat{S}^{k_0}\hat{V}\hat{S}^{k_1}\cdots\hat{V}\hat{S}^{k_n}\right) \tag{3.52}$$

Let $K=\sum_{j=0}^n k_j$. The coefficient of $(z-E_a^0)^{-1}$ (which corresponds to the residue and thus $\hat{A}^{(n)}$) occurs when the exponent $(k_0-1)+\left(\sum_{j=1}^n k_j\right)-n = K-(n+1)$ equals -1. This implies $K-(n+1)=-1$, so $K=n$. Therefore, $k_0+\cdots+k_n=n$. Hence, the expression for $\hat{A}^{(n)}$ is:

$$\hat{A}^{(n)} = -\sum_{\substack{k_0,\cdots,k_n\geq 0\\ k_0+\cdots+k_n=n}}\hat{S}^{k_0}\hat{V}\hat{S}^{k_1}\cdots\hat{V}\hat{S}^{k_n} = -\sum_{(n)}\hat{S}^{k_1}\hat{V}\hat{S}^{k_2}\cdots\hat{V}\hat{S}^{k_{n+1}}, \quad \text{where } k_1+\cdots+k_{n+1}=n \tag{3.53}$$